Negative Numbers

Exam code: 0607

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Negative numbers

What are the rules for calculations with negative numbers?

- When **multiplying** and **dividing** with negative numbers
  - Two numbers with the **same sign** make a **positive**
    - $(-12) \div (-4) = 3$
    - $(-6) \times (-4) = 24$
  - Two numbers with **different signs** make a **negative**
    - $(-12) \div 4 = -3$
    - $6 \times (-4) = -24$
- When **adding** and **subtracting** with negative numbers
  - **Subtracting a negative** number is the same as **adding** the positive
    - e.g. $5 - (-3) = 5 + 3 = 8$
  - **Adding a negative** number is the same as **subtracting** the positive
    - e.g. $7 + (-3) = 7 - 3 = 4$

Where are negative numbers used in real-life?

- **Temperature** is a common context for negative numbers
  - If the temperature is 3°C, and it cools by 5°C, the new temperature will be -2°C
    - This is equivalent to 3 - 5 = - 2
  - If the temperature is -4°C, and it warms up by 6°C, the new temperature will be 2°C
    - This is equivalent to (-4) + 6 = 2
  - To explain why (-5) - (-6) = 1, you could think of it as follows:
    - A room is -5°C, then -6°C of cold air is 'removed'
    - The room now warms to 1°C
- **Money** and **debt** is another common context for negative numbers
  - A negative sign means you owe money
  - If someone has a debt of £200, and they borrow another £400, their total debt is now £600
    - This is equivalent to (-200) + (-400) = -600

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Examiner Tips and Tricks

Your calculator isn't always as clever as you may think!

Using brackets around negative numbers will always make sure the calculator is doing what you want.

e.g. The square of negative three is $(-3) \times (-3) = 9$.
On many calculators, $-3^2 = -9$ but $(-3)^2 = 9$.
The second one is correct.

Worked Example

Complete the following table.

Calculation	Answer
3 + (-4)	
(-5) + (-8)	
7 - (-10)	
(-8) - (-6)	
(-3) × 6	
(-9) × (-2)	
(-9) ÷ (-3)	
(-10) ÷ 5	

Answer :

Calculation	Working	Answer
3 + (-4)	3 - 4	**-1**
(-5) + (-8)	(-5) - 8	**-13**
7 - (-10)	7 + 10	**17**
(-8) - (-6)	(-8) + 6	**-2**
(-3) × 6	3 × 6 = 18 one is negative	**-18**
(-9) × (-2)	9 × 2 = 18 both are negative	**18**
(-9) ÷ (-3)	9 ÷ 3 = 3 both are negative	**3**

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